
BioToken and BioFM – Biologically-Informed Tokenization Enables Accurate and Efficient Genomic Foundation Models

Aleksandr Medvedev^{*1} Karthik Viswanathan^{*123} Praveenkumar Kanithi^{*1} Kirill Vishniakov¹⁴
Prateek Munjal¹ Clement Christophe¹ Tiago Magalhaes¹ Marco AF Pimentel¹ Ronnie Rajan¹
Shadab Khan¹⁵

Abstract

Existing genomic foundation models (GFMs) typically treat DNA as raw nucleotide sequences, often overlooking the regulatory context required to interpret genetic variation accurately. We introduce BioToken, a tokenization framework that directly encodes variants and biological annotations into genomic representations, and BioFM, a parameter-efficient model built on this architecture. By leveraging biological inductive biases, BioFM outperforms state-of-the-art models and specialized baselines like Enformer on benchmarks including pathogenicity and expression prediction while requiring 100-fold less compute than current large-scale genomic models. These findings demonstrate that explicitly modeling biological structure yields more robust and efficient genomic representations than scaling alone.

1. Introduction

Genomes encode biological complexity and underpin phenotypic diversity through molecular variations. While high-throughput sequencing has improved our ability to map genetic variants associated with diseases, accurately interpreting this data remains a challenge (Marwaha et al., 2022; Cipriani et al., 2025). This difficulty arises largely from conceptual limitations in existing paradigms, which treat DNA sequences primarily as collections of nucleotides without explicit consideration of their diverse biological contexts and regulatory significance.

Drawing inspiration from Large Language Models (LLMs),

^{*}Equal contribution ¹M42, Abu Dhabi, United Arab Emirates
²Technical University of Munich, Germany ³Helmholtz Munich, Germany ⁴Ruya AI, Abu Dhabi, United Arab Emirates ⁵ADIA Lab, Abu Dhabi, United Arab Emirates. Correspondence to: Aleksandr Medvedev <amedvedev@m42.ae>.

recent research has developed Genome Foundation Models (GFMs) that apply similar architectures to genomic data (Dalla-Torre et al., 2025; Nguyen et al., 2024; Benegas et al., 2025a; Sanabria et al., 2024; Nguyen et al., 2023; Fishman et al., 2025; Shao & Yan, 2024). These models typically employ tokenization and training objectives analogous to natural language processing, such as sequential prediction or masked language modeling (Nguyen et al., 2024; Dalla-Torre et al., 2025). While effective for generative tasks (Nguyen et al., 2024; Bixi et al., 2026), this approach often overlooks regulatory annotations and cellular states that are central to genome function. Unlike natural languages, where syntactic rules are well-defined, genomic meaning is derived from context-dependent interactions and chromatin structure—areas that remain under active investigation (Gerstein et al., 2012; Kellis et al., 2014; Papp & Oliver, 2005).

GFMs must also contend with the sparsity of human genetic variation, as differences between individuals typically affect less than 1% of DNA (Consortium et al., 2015; Chen et al., 2024). Models trained with standard language objectives are incentivized to memorize common reference patterns rather than learn representations that capture the functional impact of rare variants (Tirumala et al., 2022; Kiyomaru et al., 2024). Consequently, current GFMs tend to perform well on tasks requiring only local context, such as motif discovery, but underperform on tasks requiring generalization across biological contexts, such as eQTL or variant effect prediction (Vishniakov et al., 2026). We argue that forcing models to learn genomic syntax from raw nucleotides is computationally inefficient and fails to leverage available biological structure.

To address these shortcomings, we introduce BioToken, a modular tokenization approach designed to incorporate genomic inductive biases. BioToken integrates biologically relevant annotations—including Single Nucleotide Variants (SNVs), exons, transcripts, and coding regions—directly into the DNA sequence. Unlike prior work that uses task-specific conditioning tokens (Nguyen et al., 2024) or structural tokenization (Saber et al., 2024), BioToken embeds diverse regulatory contexts during foundational pretraining.

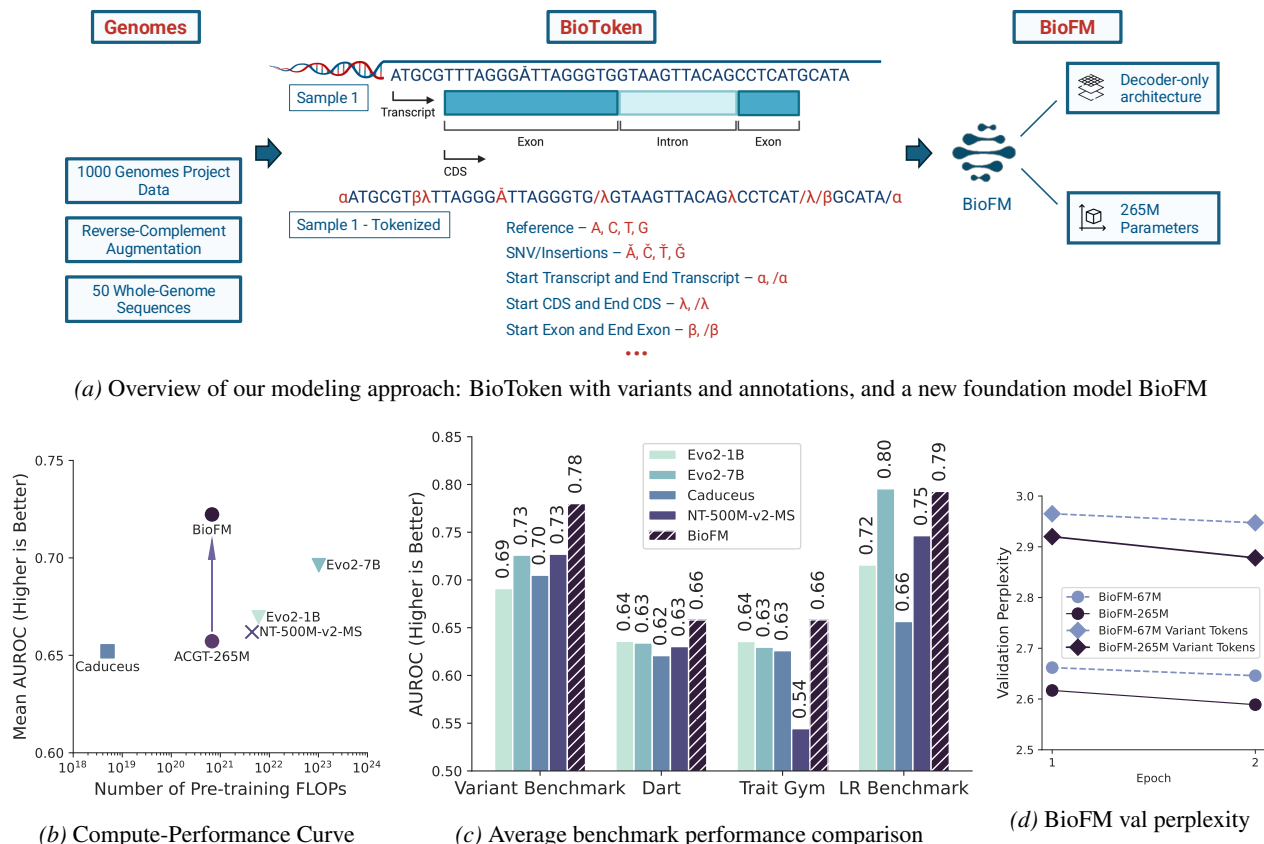


Figure 1. BioFM integrates biologically-informed tokenization, significantly outperforming GFM and specialized models across a range of tasks and benchmarks. **a)** Schematic representation of BioFM’s architecture highlighting its key components and an illustrative example showing how DNA sequences are annotated. **b)** BioFM employs next token prediction to model standard nucleotides, variants and annotations. **c)** BioFM validation perplexity decreases with increasing size and training tokens for variant and other tokens. **d)** BioFM outperforms every GFM up to 7B on three out of four variant effect benchmarks. **e)** Adding annotations to the nucleotide-only model with the same size and amount of compute (ACGT-265M) significantly increases variant effects performance across four benchmarks from (d).

Building upon this framework, we developed BioFM, a decoder-only transformer explicitly designed for computational efficiency (Vaswani et al., 2017). With 265 million parameters, BioFM is significantly smaller than contemporary models such as Nucleotide Transformer (2.5B) and Evo (40B) (Dalla-Torre et al., 2025; Nguyen et al., 2024; Brix et al., 2026).

Our contributions are as follows: (1) *BioToken*: A modular tokenization scheme that embeds biological annotations and variants directly into the sequence context, enabling models to learn from both nucleotide patterns and biological significance. (2) *BioFM*: A computational and parameter-efficient genomic foundation model (265M parameters) optimized for biological sequences. (3) *Comprehensive evaluation*: We validate BioFM using comprehensive variant benchmarks, demonstrating its utility across diverse downstream tasks.

All authors were employed by M42 at least at some point of conducting the study. M42 lead the development of BioFM,

which was among the models evaluated in this paper.

2. Methods

2.1. Biologically-Informed Tokenization of Genomes with BioToken

We developed BioToken, a modular tokenization framework design that encodes variants and biological annotations directly within DNA sequences. Conventional genomic modeling simplifies DNA sequences as strings of nucleotides, ignoring biological context essential for accurate genomic interpretation. In contrast, to help the model focus on genetic variations, we developed a tokenization strategy that replaces standard nucleotide tokens (A, G, C, T) with special variant tokens (Ä, Ğ, Č, Ĥ) to capture single nucleotide variants (SNV) and insertions. Similarly, BioToken introduces special tokens that encode genomic annotations such as transcript boundaries, exon-intron architecture, and coding DNA sequences (CDS) (Appendix A.2.1) (Figure 1a). Encoding these biologically meaningful annotations leverages

genomic inductive biases, shifting the learning objective from superficial positional memorization towards deeper functional representation learning. Notably, the modularity and extendibility of BioToken enables it to integrate additional genomic annotations not explored in this study, such as RNA secondary structures, epigenetic states (e.g., methylation, histone modifications), and chromatin accessibility patterns. BioToken therefore provides a foundation for genome-scale modeling across diverse downstream applications.

2.2. BioFM – Computationally Efficient Foundation Modeling

Using BioToken, we developed BioFM, a computationally efficient genome foundation model optimized for biologically-informed representation learning. BioFM employs a decoder-only transformer architecture comprising 265M parameters, substantially smaller than contemporary GFM, which significantly reduces computational resource demands. We implemented reverse-complement augmentation during training, randomly applying it to 50% of input sequences while applying only strand-specific annotations. BioFM was pretrained on WGS data from 50 individuals (Table 1) selected from all five superpopulations from 1000 Genomes project (Consortium et al., 2015), yielding approximately 332 billion tokens for two epoch of pretraining. We empirically determined 6K nucleotide as the ideal context window. We adopted Rotary Position Embeddings (RoPE) but adjusted the positional embedding frequency parameter θ from 10^4 to 10^5 , to more effectively model genomic dependencies over longer distances (Xiong et al., 2023). Our initial ablation analyses was conducted using a reduced-size 67M-parameter model, and all subsequent experiments were performed using a scaled-up 265M-parameter model.

3. Results

3.1. BioFM Accurately Predicts Variant Effects Across Diverse Molecular Phenotypes

We developed a new Variant Benchmark to evaluate BioFM’s ability to effectively utilize variant information across multiple biological contexts, significantly extending beyond conventional genomic benchmarks (Dalla-Torre et al., 2025; Marin et al., 2024; Grešová et al., 2023; Liu et al., 2024; Kao et al., 2024), which primarily emphasize region classification tasks. Although existing assessments such as BEND (Marin et al., 2024) and the Genomic Long-Range Benchmark (GLRB) (Kao et al., 2024) measure critical functions like noncoding pathogenicity and tissue-specific expression, they do not represent the broader spectrum of variant-mediated molecular processes. In contrast, we organize our benchmark evaluations into two major categories to systematically assess BioFM’s capabilities: (1)

Functional Effects of Genetic Variants, focusing on the direct molecular and biological consequences of genetic variation, and (2) Population-Level Variation, assessing BioFM’s sensitivity to broader genetic patterns across human populations.

3.1.1. Functional Effects of Genetic Variants

We assessed BioFM’s capability to model variant effects across five key molecular processes.

- **Coding pathogenicity assessment:** Accurate prediction of pathogenic coding variants is fundamental to precision medicine and clinical genomics. We assessed this using the AlphaMissense dataset (Cheng et al., 2023), a catalog of coding variants annotated for pathogenicity.
- **Noncoding pathogenicity assessment:** Pathogenic variants in noncoding regions significantly impact gene regulation, influencing complex traits and diseases. For evaluation, we used the BEND variant pathogenicity dataset, containing 295,000 annotated SNVs in non-coding genomic regions (Marin et al., 2024).
- **Expression effect prediction:** Variant-driven changes in gene expression contribute to phenotypic diversity and disease processes. We used the eQTL dataset from Borzoi (Linder et al., 2025) and expression dataset from DeepSea (Zhou & Troyanskaya, 2015).
- **Alternative splicing:** Variant-induced alternative splicing contributes significantly to human proteomic diversity and affects biological processes and diseases. To evaluate modeling accuracy for splicing variations across multiple cellular contexts, we used sQTL dataset from Borzoi (Linder et al., 2025).
- **DNA methylation:** Variant-driven changes in DNA methylation patterns modulate gene-environment interactions and influence developmental and disease phenotypes. To test the model’s ability to predict variant effects on DNA methylation, we utilized methylation quantitative trait loci (meQTL) data from the GRASP database (Leslie et al., 2014).

3.1.2. Population-Level Variation

We next evaluated BioFM’s ability to capture population-level insights. Specifically, we considered two tasks:

- **Local ancestry classification:** Understanding how genetic variants encode ancestral backgrounds informs studies of population structure, evolutionary biology, and disease susceptibility. Local ancestry classification tests sensitivity of GFM embeddings to small

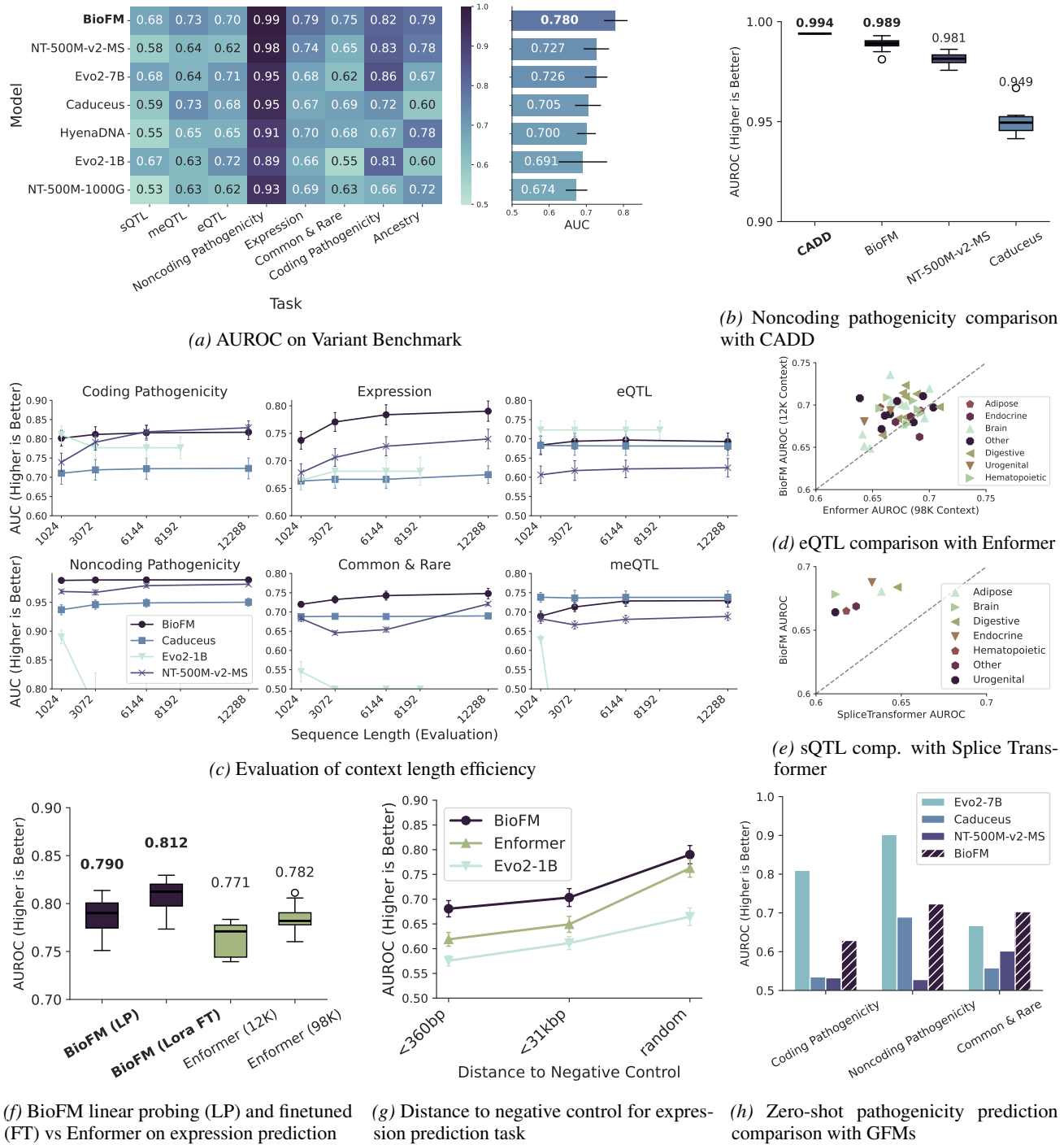


Figure 2. Extensive evaluation shows that BioFM excels at modeling variants and efficiently using genomic context for superior accuracy and generalization. Comprehensive performance evaluation of BioFM across multiple genomic prediction tasks using 11-fold cross-validation. **a)** Across diverse variant prediction tasks, BioFM outperforms other GFMs. Error bars are 2 standard deviations wide, centered on the mean. **b)** BioFM shows comparable performance to CADD for noncoding pathogenicity using linear probing. **c)** BioFM demonstrates robust scalability and accuracy across increasing context lengths, consistently outperforming competing models, particularly in noncoding pathogenicity, rare variant, and expression prediction tasks. Error bars are 2 standard deviations wide, centered on the mean. **d)** BioFM significantly outperforms Enformer on a tissue-specific eQTL dataset. **e)** BioFM achieves consistent, tissue-independent improvements over SpliceTransformer on tissue-specific splicing (sQTL) predictions. **f)** Both linear probing and finetuned BioFM improve expression prediction, exceeding Enformer even with much longer context windows. Box plot shows median, 1st and 3rd quartile, and outliers. **g)** AUROC for BioFM, Enformer, and Evo2-1B on expression prediction using negative controls at <360 bp, <31 kbp, and random variants. BioFM achieves higher performance across all settings, indicating robustness to distance-based control selection. **h)** BioFM demonstrates competitive zero-shot pathogenicity prediction, falling only behind Evo2-7B on 2 of 3 tasks.

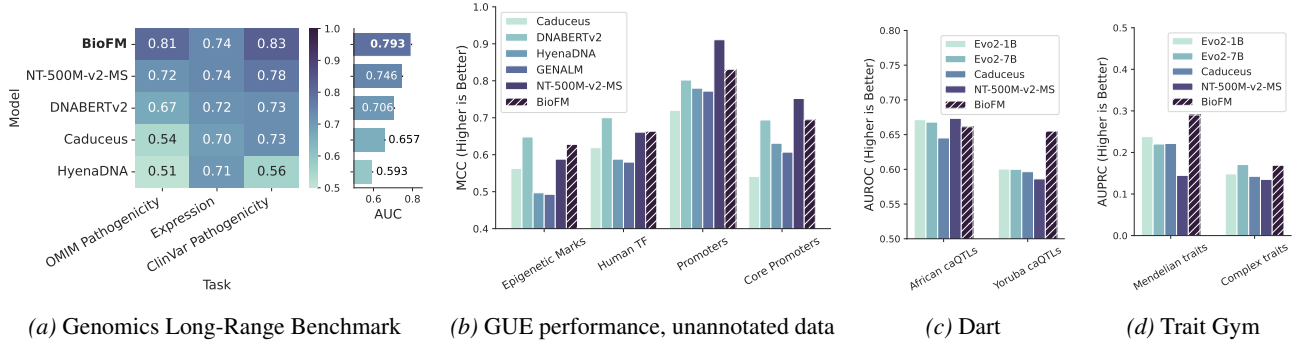


Figure 3. BioFM improves predictive accuracy on GLRB. Without BioToken, in an evaluation of worst-case performance, it achieves competitive results across genomic functional annotations on the GUE benchmark. **a)** On GLRB, BioFM outperforms previous GFM that required extensive fine-tuning (Kao et al., 2024) and longer genomic contexts. We used the same test set as (Kao et al., 2024). Results for all models except BioFM were obtained from (Kao et al., 2024). **b)** BioFM is competitive in functional genomic annotations even when specialized tokens are excluded, highlighting its generalization ability and understanding of genomic regulatory features. Every model underwent full finetuning. Detailed performance metrics are provided in Appendix, Table 5. **c)** BioFM outperforms all models on Yoruba caQTLs and is competitive on African caQTLs. We used linear probing for every model, context length is 12K except Evo2-1B and Evo2-7B, where we set it to 1K. **d)** BioFM outperforms all models on Mendelian traits task and is second-best on Complex traits task. Metric is AUPRC, following (Benegas et al., 2025b). We performed all evaluations on our codebase, matching the scheme from (Benegas et al., 2025b) as closely as possible.

changes in long sequences. We used DNA segments labeled by five major superpopulations from the 1000 Genomes (Consortium et al., 2015).

- **Common vs synthetic variants:** This task evaluates the model’s ability to recognize biologically conserved genomic contexts of authentic common variants. We randomly sampled 100K common variants (MAF > 0.05) from GnomAD (Chen et al., 2024) and paired each with a synthetic control (“rare”) variant generated by randomly substituting a nucleotide within a ± 16 -nucleotide local context window.

3.1.3. Benchmarking Protocol and Model Evaluation

We developed a standardized assessment protocol to evaluate different model architectures fairly across all tasks. For each variant, we use its DNA context up to the maximum length supported by the model, capped at 12K nucleotides. For encoder models, we center the variant within its context, using equal sequence lengths from upstream and downstream regions. For decoder-only models, we extract embeddings from upstream and downstream sequences of equal length (half the evaluation context size each), with the downstream sequence reverse complemented. We generate separate embeddings for reference and mutated sequences, average upstream and downstream embeddings for each, and concatenate the averaged reference and mutated vectors to form the final variant representation. Refer Section A.4 for full implementation details. The final variant embeddings are used as inputs to a linear classifier.

We evaluated six GFM—Evo2-7B-Base, Evo2-1B-Base, NT-500M-v2-MS, NT-500M-1000G, HyenaDNA, Ca-

duceus—on the Variant Benchmark. We selected models with different sizes, amount of compute spent ranging from 100x to 0.01x of BioFM (Figure 1b), context length from 6144 to 1M tokens. Performance for all of them is shown in Figure 2a. Additionally, we included GENA-LM, DNABERTv2, and a supervised convolutional neural network (CNN) with 120K parameters as a baseline, trained separately for each task and fold. Results for all models are shown in Figure 7.

BioFM consistently generated superior representations for variant effects prediction (overall $AUROC = 0.780$) relative to second-best NT-500M-v2-MS ($AUROC = 0.727$), despite nearly half the parameter count Figure 2a. BioFM was able to outperform Evo2-7B which used approximately 100x more compute during pretraining on five out of eight tasks. Notably, BioFM significantly outperformed NT-500M-1000G, highlighting limitations of fixed-length k-mer tokenization for variant-rich data. BioFM achieved the highest performance across six of eight tasks, with significant advantages particularly in distinguishing common versus rare variants ($AUROC = 0.746$ vs. 0.69) and expression prediction ($AUROC = 0.786$ vs. 0.74).

3.1.4. Context Length Utilization and Embedding Quality

In general, transformer based GFM are known to struggle to fully utilize their context windows (Kuratov et al., 2024). To investigate how feature extraction is affected by context length, we evaluated model performance using variant contexts ranging from 1K to 12K nucleotides (Figure 2c).

BioFM demonstrates strong performance even with a 1K-nucleotide context, significantly outperforming NT-

500M-v2-MS across all tasks, including coding pathogenicity. However, NT-500M-v2-MS scales slightly better for pathogenicity tasks. For expression and rare variants prediction, BioFM’s performance does not saturate, suggesting potential benefits from models with even longer context windows. BioFM effectively utilizes its full 6K-nucleotide window (6K in forward strand and 6K in reverse strand). By contrast, Caduceus, despite being designed for long contexts, plateaued around 3K nucleotides across all tasks. Based on these findings, we limited evaluation contexts to a maximum of 12K nucleotides. Surprisingly, the performance of larger models such as Evo2-1B and Evo2-7B decays after 1K nucleotides for four out of six tasks. This is likely due to mean pooling strategy for extracting features. Across all context lengths, BioFM was the only model consistently ranking among the top two performers.

3.1.5. Direct Comparison with Specialized Models

We compared BioFM to strong supervised genomic baselines: Enformer and Borzoi for expression and tissue-specific eQTL prediction, and Splice Transformer for sQTL prediction. Since both models produce tissue- or cell type-specific outputs that do not always match with our data tracks, we extracted their sequence embeddings for downstream linear probing to ensure a fair comparison. As shown in Figure 2f, BioFM matches Enformer’s performance on expression prediction when both use 12K context, making it the first GFM to do so. We then finetuned BioFM with parameter-efficient LoRA on the expression prediction dataset and compared it with Enformer using 98K context (Figure 2f). BioFM significantly outperformed Enformer, median $AUROC = 0.812$ vs $AUROC = 0.782$ ($p = 0.002$ subsection A.5). BioFM with linear probing also outperforms Enformer on eQTL dataset from Borzoi (Linder et al., 2025), $AUROC = 0.697$ vs $AUROC = 0.674$ (Figure 2d). Moreover, BioFM is significantly better than Splice Transformer for sQTL prediction (Figure 2e), achieving median tissue $AUROC = 0.676$ compared to $AUROC = 0.629$ of Splice Transformer. BioFM also beats Borzoi, achieving eQTL $AUROC = 0.699$ compared to Borzoi’s $AUROC = 0.689$, and sQTL $AUROC = 0.683$ vs Borzoi’s $AUROC = 0.63$. Analyzing tissue-cluster-specific performance (Figure 2e) reveals that BioFM consistently performs better across all tissue clusters. However, BioFM is not able to outperform CADD on noncoding pathogenicity prediction, albeit it is better than other GFMs (Figure 2b).

3.1.6. Computational Efficiency and Generalization Compared to Large Models

For many models, the likelihood of a nucleotide at a given position is correlated with its pathogenicity. To evaluate the zero-shot performance of BioFM, we compared it with Evo2-7B (Brix et al., 2026), which was trained on 3T nu-

cleotide tokens spanning all domains of life. We restricted the context size to 1K nucleotides, as Evo2 models exhibit reduced performance at longer sequences, a trend observed in our experiments (Figure 2c) and also reported in the original Evo2 paper (Brix et al., 2026).

On pathogenicity prediction tasks, Evo2-7B achieved higher accuracy, but it performed worse on the Common & Rare classification task. In contrast, BioFM produced competitive results while being approximately two orders of magnitude more compute-efficient in generating variant representations (Figure 2a). This efficiency stems from its use of variant tokens and annotations, which allow the model to encode variation explicitly rather than relying solely on scale. The performance boost on the Common & Rare task can be attributed to both the incorporation of variant tokens, as demonstrated in our ablation studies (Figure 5a), and training on 1000G data. Importantly, this task is particularly challenging because common and rare variants were deliberately chosen to be in close proximity within small genomic regions, requiring models to be sensitive to subtle, local nucleotide frequency shifts.

In conclusion, these findings establish BioFM as a meaningful advance in genomic modeling - it is currently the only genomic foundation model that demonstrates practical utility for both variant expression and splicing effects, matching or exceeding the performance of specialized models designed specifically for these tasks.

3.1.7. BIOFM PERFORMS WELL ON GLRB, DART, TRAIT GYM BENCHMARKS

We evaluated BioFM on three additional benchmarks: the Genomic Long-Range Benchmark (GLRB) (Kao et al., 2024), Dart-Eval (Patel et al., 2024), and Trait Gym (Bene-gas et al., 2025b). GLRB includes noncoding pathogenicity, expression, and OMIM-versus-common variant prediction tasks, though its datasets were preprocessed differently from those in our Variant Benchmark. Dart focuses on dsQTLs in African and caQTLs in Yoruba populations, representing molecular phenotypes not covered in Variant Benchmark. Trait Gym assesses the detection of variants influencing Mendelian and complex traits, simulating GWAS-style inference.

BioFM outperforms NT-500M-v2-MS and other models on GLRB ($AUROC = 0.793$), despite NT-500M-v2-MS being finetuned for extended context lengths (12K to 66K nucleotides, Figure 3a). This suggests that inductive biases introduced through BioToken are more effective than finetuning or context extension alone, and that GLRB tasks may not strongly depend on long-range interactions. On Dart, BioFM performs significantly better on Yoruba caQTLs, though not on the more diverse African dsQTL dataset (Figure 3c), likely reflecting both the explicit use of variant to-

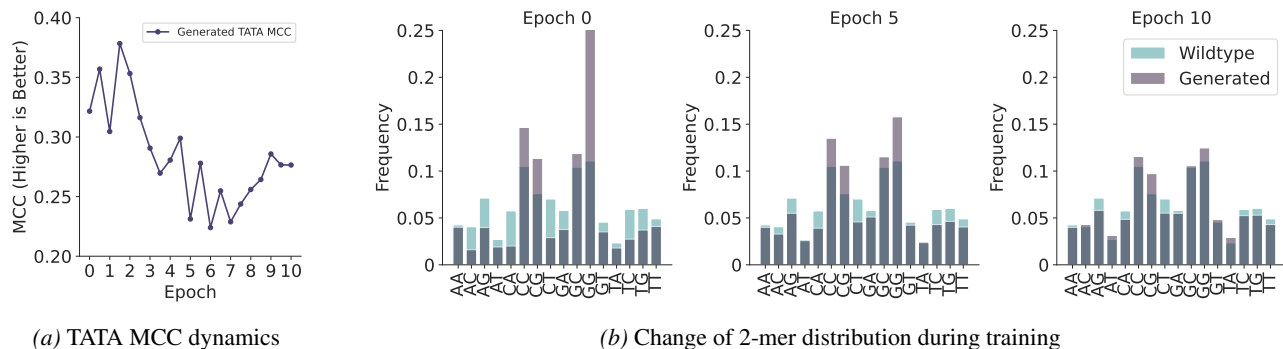


Figure 4. BioFM effectively generates biologically plausible synthetic promoter sequences, progressively aligning with naturally occurring promoter features through fine-tuning. **a)** MCC tracking of TATA prediction accuracy across 10 training epochs. Performance is evaluated based on the model’s ability to correctly generate TATA-box-containing promoters for genes naturally possessing wildtype TATA-box promoters (true positives) while avoiding incorrect TATA-box insertions in naturally non-TATA promoters (false positives). **b)** Comparative analysis of 2-mer frequency distributions between generated and wildtype promoter sequences, demonstrating progressive convergence of synthetic promoter sequence composition toward natural distributions during the finetuning process.

kens and the presence of two Yoruba samples in pretraining. On TraitGym, BioFM shows strong performance in identifying variants underlying Mendelian traits (Figure 3d), showing the importance of biological annotations and variant tokens. Overall, across GLRB, Dart, and Trait Gym, BioFM outperforms longer-context models such as Caduceus and larger models such as Evo2-7B, underscoring the value of variant-aware tokenization and genomic pretraining.

3.1.8. BIOFM CAPTURES BROAD FUNCTIONAL GENOMIC SIGNALS

To assess BioFM’s general applicability as a foundation model, especially in scenarios where annotations are not available (and therefore annotation tokens are not included as part of the input), we evaluated its performance on the Genomic Understanding Evaluation (GUE) benchmark (Grešová et al., 2023). GUE benchmark includes prediction of epigenetic modifications, transcription factor binding sites, and promoter region characteristics, structured as binary region classification problems analogous to those in the NT benchmark (Dalla-Torre et al., 2025). Notably, these evaluations were conducted without BioFM’s specialized features (BioToken’s annotation tokens, variant tokens were not used, no reverse complement augmentation), resulting in a lower-bound result for BioFM.

BioFM demonstrated consistently strong performance across all task categories within the GUE benchmark, ranking second overall. It was surpassed only by DNABERTv2 on epigenetic mark and transcription factor binding predictions, and by NT-500M-v2-MS on promoter-specific tasks. BioFM consistently outperformed competing models such as HyenaDNA, Caduceus, and GENA-LM across all evaluated categories. Further, BioFM surpassed NT-500M-v2-MS on epigenetic mark and transcription factor binding

tasks. These results demonstrate that BioFM is useful even in applications where annotations are not available.

3.2. INSTRUCTION-TUNING ENABLES BIOFM TO GENERATE REGULATORY SEQUENCES

The generation of synthetic promoters tests BioFM’s generative capabilities. The ability to design promoters enables precise control over transcription, with applications in optimizing gene expression for research and therapeutic interventions (Mohamed et al., 2016; Tominaga et al., 2024). As promoters are typically located upstream of the gene, BioFM requires fine-tuning to generate them effectively. We prompt model with a gene sequence and a special token ‘P’ after it, creating instruction set of $\langle \text{DNA sequence} \rangle \langle \text{Task-identifier token} \rangle \langle \text{Output} \rangle$. Training was done using instruction-answer pair of genes-promoters set and testing on a held-out set. Promoter sequences were drawn from the EPDNew dataset (Meylan et al., 2020)

We systematically validated BioFM-generated promoters using three independent biological criteria: (1) compositional similarity to natural promoter sequences, and (2) accurate modeling of biologically significant promoter types (TATA versus non-TATA).

First, we assessed compositional similarity by analyzing k-mer distributions of generated promoter sequences. We found that after 10 epochs of fine-tuning, BioFM-generated promoters successfully captured the k-mer distributions of wildtype promoters, indicating convergence toward biologically realistic promoter compositions (Figure 4b). Second, we measured BioFM’s ability to model biologically significant promoter types accurately. The Matthews Correlation Coefficient (MCC) for predicting TATA-containing promoters was 0.28 after fine-tuning (Figure 4a), confirming the

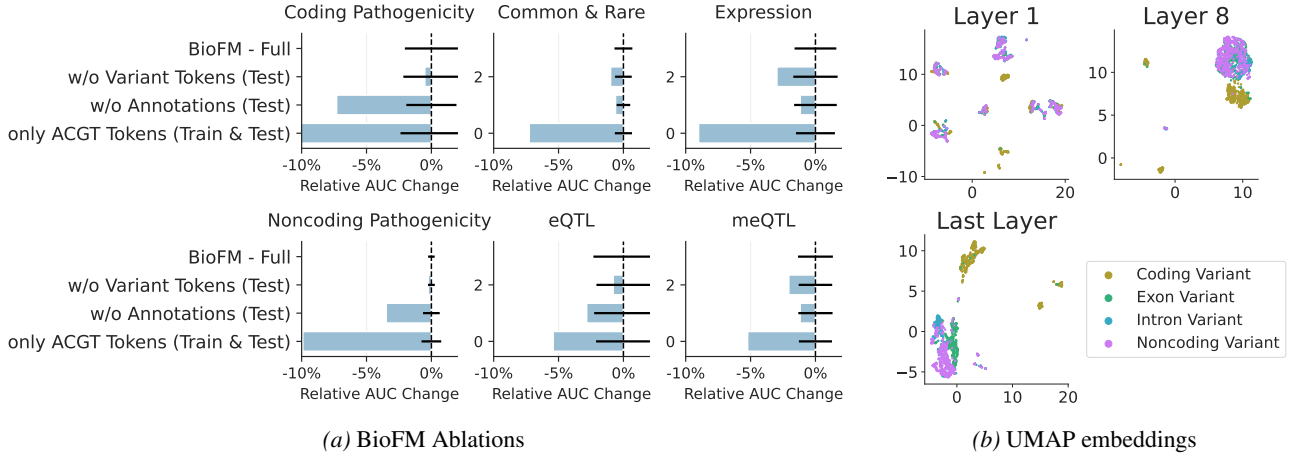


Figure 5. Detailed analysis shows that biologically-informed tokenization using BioToken in BioFM significantly enhances accuracy. **a)** Ablation experiments show that both annotation and variant-specific tokens in BioToken significantly contribute to improved performance, specifically for coding pathogenicity and splicing prediction. We removed annotations and variant tokens from test sequences and tested the original BioFM. We trained a version of BioFM without annotation tokens and another version without annotation and variant tokens. Error bars are 2 SD wide, centered on the mean. **b)** UMAP embeddings illustrate progressive differentiation of variant types through model layers, indicating BioFM effectively learns biologically meaningful distinctions such as coding vs noncoding variants.

model’s capability in discriminating genes more likely to possess TATA promoters. The median similarity of generated promoters is 0.44, which shows that BioFM can generate diverse sequences.

3.3. Ablation, Scaling, Representation Analysis, and Attention Specialization

First, we quantified the impact of BioToken through ablation experiments, where we ablated the variant and annotation tokens (Figure 5a). We established baseline comparisons by training a nucleotide-only model (ACGT-265M) on the same subset from 1000G, without any specialized tokens. Then, we assessed the impact of annotation tokens by training an intermediate model that included variant tokens but excluded annotations. During evaluation, we selectively removed annotation and/or variant tokens from test sequences to show that both variant and annotation tokens significantly improve BioFM performance. Annotation tokens are particularly important for pathogenicity prediction, while variant tokens have notable impact on rare variant, expression, and meQTL tasks.

We also tested if other models can be improved during evaluation by concatenating simple annotations feature vector with resulting embedding. Specifically, we added count of CDS, exon, transcript annotations in 128 or 12K nucleotides closest to the variant to the variant embedding, and used XGBoost instead of linear probing for Variant Benchmark Table 9. BioFM only modestly improved from $AUROC = 0.780$ to 0.786 . NT-500M-v2-MS improved from 0.720 to 0.748 , while Caduceus is the only model with notable improvement from 0.719 to 0.777 . It shows

that BioFM embeddings include annotation-related biological context already, while Caduceus can benefit a lot from including annotation in pretraining stage. More detailed discussion is in in subsection A.6 and subsection A.8.

Next, we investigated the structure of variant representations learned by BioFM using UMAP embeddings derived from pathogenicity benchmark datasets (Figure 5b). This embedding analysis revealed progressive differentiation of variants in deeper model layers. While initial layers showed no biologically coherent separation, the final layer clearly distinguished coding from noncoding variants, with further sub-structuring between exonic and intronic variants. We also discovered that BioFM last layer consistently produces the best embeddings for all tasks Figure 6, unlike NT and Evo2 model families, where middle layers usually show better performance.

4. Discussion and Conclusion

In this work, we introduced BioToken, a biologically-informed tokenization framework, and BioFM, a BioToken-based model which improves genomic representations. Unlike other GFMs, BioFM explicitly encodes variants and genomic annotations with specialized tokens. It enables BioFM to surpass even larger GFMs, such as Evo2-7B, across multiple tasks. BioFM matches or exceeds the performance of specialized models like Enformer and Splice-Transformer on domain-specific benchmarks. These results demonstrate the model’s capability to capture variant effects across the central dogma, perform region classification, and generate promoter sequences.

The primary advantage of BioToken is the shift from nucleotide-only processing to a representation that integrates biological context. Ablation studies confirm that embedding annotations, such as SNVs, exons, and transcripts, is the key factor driving the improved performance. The framework is modular and extensible; future iterations could incorporate chromatin structure, epigenetic markers, or representations from single-cell foundation models (Yang et al., 2022; Cui et al., 2024a;b; Fu et al., 2025; Team, 2024). This approach is also adaptable to protein language models, offering a path to unified multi-modal representations. Integrating biological inductive biases also greatly improves computational efficiency. By removing implicit learning of genomic structure, BioFM achieves competitive results using 333 billion training tokens, significantly fewer than the trillions employed in recent large-scale studies (Dalla-Torre et al., 2025; Brixi et al., 2026).

We acknowledge distinct limitations. First, BioToken relies on high-quality annotations, which may restrict its application to non-model organisms with limited genomic resources. Second, while trained on data from 50 individuals, the scaling properties of the model with larger cohorts, such as biobank-scale data (Träuble et al., 2025), remain to be explored. Finally, while BioFM is still competitive in the no-annotation setting, as was shown on GUE benchmark, its utility in expanding annotations (e.g. CDS, exon detection) is underexplored.

Also, adding annotation and variant tokens during pretraining can raise concerns about bias during evaluation. For example, variants closer to transcript boundaries are more likely to influence its expression. To partially alleviate them, we additionally tested BioFM on expression prediction with distance-matched negative controls. Results indicate that BioFM maintains robust performance (Figure 2g). This suggests that explicit annotation tokens facilitate effective feature prioritization rather than solely relying on positional correlations.

In conclusion, BioToken and BioFM provide a robust framework for modeling the complex relationships between genetic variation and molecular phenotypes. By prioritizing biological context and computational efficiency, this work offers a scalable paradigm for advancing computational genomics.

Software and Data

We provide a code to generate BioFM embeddings for variants of interest here: <https://github.com/m42-health/biofm-eval>. Variant Benchmark dataset is on huggingface, <https://huggingface.co/datasets/m42-health/variant-benchmark>.

Impact Statement

This paper presents work aimed at advancing the field of machine learning for genomics, with the goal of improving the interpretation of genetic variation and its link to phenotypic traits. By introducing a framework that achieves state-of-the-art performance with significantly reduced computational requirements, our approach lowers the barrier to entry for genomic research and reduces the environmental carbon footprint associated with training large-scale models. This efficiency facilitates the democratization of advanced genomic analysis, allowing execution on consumer-grade hardware.

However, we acknowledge that the performance of BioToken and BioFM relies on the quality and completeness of biological annotations. Biases inherent in existing reference genomes or annotation databases, particularly regarding the representation of diverse ancestral populations, may propagate to model predictions. Consequently, care must be taken to ensure that such biases are identified and mitigated before deploying these models in diverse populations. Furthermore, while our model demonstrates improved capability in predicting variant pathogenicity, it is intended as a research tool. It should not be used for clinical diagnosis or medical decision-making without further rigorous validation and regulatory compliance.

References

- Avsec, Ž., Agarwal, V., Visentin, D., Ledsam, J. R., Grabska-Barwinska, A., Taylor, K. R., Assael, Y., Jumper, J., Kohli, P., and Kelley, D. R. Effective gene expression prediction from sequence by integrating long-range interactions. *Nature methods*, 2021.
- Benegas, G., Albors, C., Aw, A. J., Ye, C., and Song, Y. S. A dna language model based on multispecies alignment predicts the effects of genome-wide variants. *Nature Biotechnology*, pp. 1–6, 2025a.
- Benegas, G., Eraslan, G., and Song, Y. S. Benchmarking dna sequence models for causal regulatory variant prediction in human genetics. *bioRxiv*, 2025b.
- Brixi, G., Durrant, M. G., Ku, J., Naghipourfar, M., Poli, M., Sun, G., Brockman, G., Chang, D., Fanton, A., Gonzalez, G. A., et al. Genome modelling and design across all domains of life with Evo 2. *Nature*, 652(8112):1349–1361, 2026.
- Chen, S., Francioli, L. C., Goodrich, J. K., Collins, R. L., Kanai, M., Wang, Q., Alföldi, J., Watts, N. A., Vittal, C., Gauthier, L. D., et al. A genomic mutational constraint map using variation in 76,156 human genomes. *Nature*, 625(7993):92–100, 2024.

- Cheng, J., Novati, G., Pan, J., Bycroft, C., Žemgulytė, A., Applebaum, T., Pritzel, A., Wong, L. H., Zielinski, M., Sargeant, T., et al. Accurate proteome-wide missense variant effect prediction with alphamissense. *Science*, 381(6664):eadg7492, 2023.
- Cipriani, V., Vestito, L., Magavern, E. F., Jacobsen, J. O., Arno, G., Behr, E. R., Benson, K. A., Bertoli, M., Bockenhauer, D., Bowl, M. R., et al. Rare disease gene association discovery in the 100,000 genomes project. *Nature*, pp. 1–9, 2025.
- Consortium, . G. P. et al. A global reference for human genetic variation. *Nature*, 2015.
- Cui, H., Wang, C., Maan, H., Pang, K., Luo, F., Duan, N., and Wang, B. scgpt: toward building a foundation model for single-cell multi-omics using generative ai. *Nature Methods*, 21(8):1470–1480, 2024a.
- Cui, Z., Xu, T., Wang, J., Liao, Y., and Wang, Y. Geneformer: Learned gene compression using transformer-based context modeling. In *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 8035–8039. IEEE, 2024b.
- Dalla-Torre, H., Gonzalez, L., Mendoza-Revilla, J., Lopez Carranza, N., Grzywaczewski, A. H., Oteri, F., Dallago, C., Trop, E., de Almeida, B. P., Sirelkhatim, H., et al. Nucleotide transformer: building and evaluating robust foundation models for human genomics. *Nature Methods*, 22(2):287–297, 2025.
- Fishman, V., Kuratov, Y., Shmelev, A., Petrov, M., Penzar, D., Shepelin, D., Chekanov, N., Kardymon, O., and Burtsev, M. GENA-LM: a family of open-source foundational DNA language models for long sequences. *Nucleic Acids Research*, 53(2):gkae1310, 2025.
- Frazer, J., Notin, P., Dias, M., Gomez, A., Min, J. K., Brock, K., Gal, Y., and Marks, D. S. Disease variant prediction with deep generative models of evolutionary data. *Nature*, 599(7883):91–95, 2021.
- Fu, X., Mo, S., Buendia, A., Laurent, A. P., Shao, A., Alvarez-Torres, M. d. M., Yu, T., Tan, J., Su, J., Sagatelian, R., et al. A foundation model of transcription across human cell types. *Nature*, pp. 1–9, 2025.
- Gerstein, M. B., Kundaje, A., Hariharan, M., Landt, S. G., Yan, K.-K., Cheng, C., Mu, X. J., Khurana, E., Rozowsky, J., Alexander, R., et al. Architecture of the human regulatory network derived from encode data. *Nature*, 489(7414):91–100, 2012.
- Grešová, K., Martinek, V., Čechák, D., Šimeček, P., and Alexiou, P. Genomic benchmarks: a collection of datasets for genomic sequence classification. *BMC Genomic Data*, 2023.
- Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., Casas, D. d. l., Bressand, F., Lengyel, G., Lample, G., Saulnier, L., et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.
- Kao, C. H., Trop, E., Polen, M., Schiff, Y., de Almeida, B. P., Gokaslan, A., Pierrot, T., and Kuleshov, V. Advancing dna language models: The genomics long-range benchmark. In *ICLR 2024 Workshop on Machine Learning for Genomics Explorations*. ICLR, 1 2024.
- Kellis, M., Wold, B., Snyder, M. P., Bernstein, B. E., Kundaje, A., Marinov, G. K., Ward, L. D., Birney, E., Crawford, G. E., Dekker, J., et al. Defining functional dna elements in the human genome. *Proceedings of the National Academy of Sciences*, 111(17):6131–6138, 2014.
- Kiyomaru, H., Sugiura, I., Kawahara, D., and Kurohashi, S. A comprehensive analysis of memorization in large language models. In *Proceedings of the 17th International Natural Language Generation Conference*, pp. 584–596, 2024.
- Kuratov, Y., Bulatov, A., Anokhin, P., Rodkin, I., Sorokin, D., Sorokin, A., and Burtsev, M. Babilong: Testing the limits of llms with long context reasoning-in-a-haystack. *arXiv preprint arXiv:2406.10149*, 2024.
- Landrum, M. J., Chitipiralla, S., Brown, G. R., Chen, C., Gu, B., Hart, J., Hoffman, D., Jang, W., Kaur, K., Liu, C., et al. Clinvar: improvements to accessing data. *Nucleic acids research*, 48(D1):D835–D844, 2020.
- Leslie, R., O’Donnell, C. J., and Johnson, A. D. Grasp: analysis of genotype–phenotype results from 1390 genome-wide association studies and corresponding open access database. *Bioinformatics*, 30(12):i185–i194, 2014.
- Linder, J., Srivastava, D., Yuan, H., Agarwal, V., and Kelley, D. R. Predicting rna-seq coverage from dna sequence as a unifying model of gene regulation. *Nature Genetics*, pp. 1–13, 2025.
- Liu, Z., Li, J., Li, S., Zang, Z., Tan, C., Huang, Y., Bai, Y., and Li, S. Z. Genbench: A benchmarking suite for systematic evaluation of genomic foundation models. *arXiv preprint arXiv:2406.01627*, 2024.
- Marin, F. I., Teufel, F., Horrender, M., Madsen, D., Pultz, D., Winther, O., and Boomsma, W. Bend: Benchmarking dna language models on biologically meaningful tasks. *ICLR*, 2024.
- Marwaha, S., Knowles, J. W., and Ashley, E. A. A guide for the diagnosis of rare and undiagnosed disease: beyond the exome. *Genome medicine*, 14(1):23, 2022.

- McLaren, W. et al. The ensembl variant effect predictor genome biology 17, 122, 2016.
- Meylan, P., Dreos, R., Ambrosini, G., Groux, R., and Bucher, P. Epd in 2020: enhanced data visualization and extension to ncna promoters. *Nucleic acids research*, 48(D1):D65–D69, 2020.
- Mohamed, H., Chernajovsky, Y., and Gould, D. Assembly pcr synthesis of optimally designed, compact, multi-responsive promoters suited to gene therapy application. *Scientific reports*, 6(1):29388, 2016.
- Mudge, J. M., Carbonell-Sala, S., Diekhans, M., Martinez, J. G., Hunt, T., Jungreis, I., Loveland, J. E., Arnan, C., Barnes, I., Bennett, R., Berry, A., Bignell, A., Cerdán-Vélez, D., Cochran, K., Cortés, L. T., Davidson, C., Donaldson, S., Dursun, C., Fatima, R., Hardy, M., Hebbar, P., Hollis, Z., James, B. T., Jiang, Y., Johnson, R., Kaur, G., Kay, M., Mangan, R. J., Maquedano, M., Gómez, L. M., Mathlouthi, N., Merritt, R., Ni, P., Palumbo, E., Perteghella, T., Pozo, F., Raj, S., Sisu, C., Steed, E., Sumathipala, D., Suner, M.-M., Uszczynska-Ratajczak, B., Wass, E., Yang, Y. T., Zhang, D., Finn, R. D., Gerstein, M., Guigó, R., Hubbard, T. J. P., Kellis, M., Kundaje, A., Paten, B., Tress, M. L., Birney, E., Martin, F. J., and Frankish, A. Gencode 2025: reference gene annotation for human and mouse. *Nucleic acids research*, January 2025.
- Nguyen, E., Poli, M., Faizi, M., Thomas, A. W., Wornow, M., Birch-Sykes, C., Massaroli, S., Patel, A., Rabideau, C. M., Bengio, Y., Ermon, S., Re, C., and Baccus, S. HyenaDNA: Long-range genomic sequence modeling at single nucleotide resolution. In *NeurIPS*, 12 2023.
- Nguyen, E., Poli, M., Durrant, M. G., Kang, B., Katrekar, D., Li, D. B., Bartie, L. J., Thomas, A. W., King, S. H., Bixi, G., Sullivan, J., Ng, M. Y., Lewis, A., Lou, A., Ermon, S., Baccus, S. A., Hernandez-Boussard, T., Ré, C., Hsu, P. D., and Hie, B. L. Sequence modeling and design from molecular to genome scale with evo. *Science*, 2024.
- Papp, B. and Oliver, S. Genome-wide analysis of the context-dependence of regulatory networks. *Genome biology*, 6(2):206, 2005.
- Patel, A., Singhal, A., Wang, A., Pampari, A., Kasowski, M., and Kundaje, A. Dart-eval: A comprehensive dna language model evaluation benchmark on regulatory dna. *Advances in Neural Information Processing Systems*, 37: 62024–62061, 2024.
- Rentzsch, P., Witten, D., Cooper, G. M., Shendure, J., and Kircher, M. Cadd: predicting the deleteriousness of variants throughout the human genome. *Nucleic acids research*, 47(D1):D886–D894, 2019.
- Saberi, A., Choi, B., Wang, S., Hernández-Corchado, A., Naghipourfar, M., Namini, A. M., Ramani, V., Emad, A., Najafabadi, H. S., and Goodarzi, H. A long-context rna foundation model for predicting transcriptome architecture. *BioRxiv*, pp. 2024–08, 2024.
- Sanabria, M., Hirsch, J., Joubert, P. M., and Poetsch, A. R. Dna language model grover learns sequence context in the human genome. *Nature Machine Intelligence*, 6(8): 911–923, 2024.
- Shao, B. and Yan, J. A long-context language model for deciphering and generating bacteriophage genomes. *Nature Communications*, 15(1):9392, 2024.
- Team, C. Chameleon: Mixed-modal early-fusion foundation models. *arXiv preprint arXiv:2405.09818*, 2024.
- Tirumala, K., Markosyan, A., Zettlemoyer, L., and Aghajanyan, A. Memorization without overfitting: Analyzing the training dynamics of large language models. *Advances in Neural Information Processing Systems*, 35: 38274–38290, 2022.
- Tominaga, M., Shima, Y., Nozaki, K., Ito, Y., Someda, M., Shoya, Y., Hashii, N., Obata, C., Matsumoto-Kitano, M., Suematsu, K., et al. Designing strong inducible synthetic promoters in yeasts. *Nature Communications*, 15(1):10653, 2024.
- Träuble, F., Stuart, L., Georgiou, A., Notin, P., Mehrjou, A., Schwessinger, R., Chevalley, M., Branson, K., Schölkopf, B., van Duijn, C., Marks, D., and Schwab, P. Multi-megabase scale genome interpretation with genetic language models, 2025. URL <https://arxiv.org/abs/2501.07737>.
- Vaswani, A., Shazeer, N. M., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. Attention is all you need. In *NeurIPS*, 2017.
- Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., Carey, C. J., Polat, İ., Feng, Y., Moore, E. W., VanderPlas, J., Laxalde, D., Perktold, J., Cimrman, R., Henriksen, I., Quintero, E. A., Harris, C. R., Archibald, A. M., Ribeiro, A. H., Pedregosa, F., van Mulbregt, P., and SciPy 1.0 Contributors. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.
- Vishniakov, K., Viswanathan, K., Medvedev, A., Kanithi, P., Pimentel, M. A., Rajan, R., and Khan, S. Tokenization to transfer: Do genomic foundation models learn good

- representations? In *The Fourteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=4UY1NHG5Ge>.
- von Werra, L., Belkada, Y., Tunstall, L., Beeching, E., Thrush, T., Lambert, N., Huang, S., Rasul, K., and Gallouédec, Q. Trl: Transformer reinforcement learning. <https://github.com/huggingface/trl>, 2020.
- Wang, Q. S., Kelley, D. R., Ulirsch, J., Kanai, M., Sadhuka, S., Cui, R., Albors, C., Cheng, N., Okada, Y., et al. Leveraging supervised learning for functionally informed fine-mapping of cis-eqtls identifies an additional 20,913 putative causal eqtls. *Nature Communications*, 12(1): 3394, 2021.
- Xiong, W., Liu, J., Molybog, I., Zhang, H., Bhargava, P., Hou, R., Martin, L., Rungta, R., Sankararaman, K. A., Oguz, B., et al. Effective long-context scaling of foundation models. *arXiv preprint arXiv:2309.16039*, 2023.
- Yang, F., Wang, W., Wang, F., Fang, Y., Tang, D., Huang, J., Lu, H., and Yao, J. scbert as a large-scale pretrained deep language model for cell type annotation of single-cell rna-seq data. *Nature Machine Intelligence*, 4(10):852–866, 2022.
- Zhou, J. and Troyanskaya, O. G. Predicting effects of noncoding variants with deep learning-based sequence model. *Nature methods*, 2015.
- Zhou, Z., Ji, Y., Li, W., Dutta, P., Davuluri, R. V., and Liu, H. DNABERT-2: Efficient foundation model and benchmark for multi-species genomes. *ICLR*, 2024.

A. Methods

A.1. Data

A.1.1. PRETRAINING DATASET

We utilized whole genome sequencing (WGS) data from the 1000 Genomes Project ([Consortium et al., 2015](#)) as the pretraining dataset. From the total pool of 3,202 available samples, we selected a subset of 50 individuals. This subset was balanced across the five superpopulations defined in the 1000 Genomes dataset, with 10 individuals randomly chosen from each superpopulation. The list of individuals is in table [Table 1](#). We used Phase 3 WGS data in HG38 assembly. Variants are phased as described in ([Consortium et al., 2015](#)).

During training, we generated input sequences by sampling genomic regions of 6,144 nucleotides in length. To increase data diversity and ensure overlap between sequences, we introduced a sliding window approach with an overlap of 1,000 nucleotides between consecutive regions. Additionally, for each sequence, we applied a random start shift uniformly sampled between 0 and 1,000 nucleotides to further vary the input segments. As data is phased, each time we randomly selected one of two haplotypes.

We excluded any 6,144-nucleotide region in which more than 50% of the bases were unknown (i.e., represented by 'N') in the human reference genome, to ensure data quality. For the remaining sequences, we introduced sample-specific genetic variation. Specifically, for each selected region, we randomly chose one of the 50 individuals and applied all of their corresponding single nucleotide variants (SNVs) and insertions or deletions (indels) that mapped to that region. This process was expected to yield a mutated sequence reflecting real human genetic diversity. The training process covers each region for each individual exactly twice. The total number of regions is 542,688, each of them is seen by the model 100 times, yielding total number of tokens equal to 333B.

To standardize input length for the model, each mutated sequence was truncated or padded to exactly 6,144 tokens. We further applied reverse complement augmentation with a probability of 50% for each region. When reverse complementing, transcript, exon, and coding sequence boundary annotations were flipped only if the original strand corresponded to the forward strand; otherwise, they were left unchanged to preserve correct strand-specific annotation. Pretraining was performed using data from autosomes (chromosomes 1 to 22) and the X chromosome.

AFR	EUR	SAS	EAS	AMR
NA19835	NA20506	NA20902	HG02040	HG01281
HG03571	HG00251	HG04001	HG01808	HG02312
NA19258	HG01707	HG03899	NA18637	HG00639
HG03055	HG00365	HG03905	HG01858	HG01374
NA18930	NA12872	HG02662	HG01878	HG01950
NA19320	NA12812	HG03703	HG00706	HG01974
NA19035	HG00159	HG04176	HG00472	HG01178
HG02307	HG00272	HG02649	HG01597	HG02285
HG03086	NA12275	HG03885	NA18608	HG02292
NA19379	HG01771	HG03778	HG00694	HG01200

Table 1. 50 samples used for pretraining BioFM

A.1.2. EVALUATION DATASETS

Noncoding Pathogenicity: This task involves classifying genetic variants located in noncoding regions of the human genome as either pathogenic (disease-causing) or benign (non-disease-causing). We use the BEND benchmark dataset ([Marin et al., 2024](#)), derived from ClinVar ([Landrum et al., 2020](#)) and includes single nucleotide polymorphisms (SNPs) annotated with binary pathogenicity labels. Specifically, the dataset comprises 295,923 SNPs, of which 21,524 are labeled as pathogenic and 274,399 as benign, resulting in a significant class imbalance (5% pathogenic). All variants are mapped to the GRCh38 reference genome.

To construct BEND, the authors first filtered ClinVar (variant summary file dated 2023-07-02) for single nucleotide variants with at least one-star review status. To isolate noncoding SNPs, they removed variants annotated as coding in ClinVar, and further excluded SNPs located in coding sequences (CDS), start codons, and stop codons based on GENCODE v43

annotations (Frazer et al., 2021). Variants from the mitochondrial genome were also omitted to maintain consistency with prior literature baselines. Pathogenicity annotations were binarized by combining the "Likely pathogenic" and "Pathogenic" categories into a single pathogenic class, and "Likely benign" and "Benign" into a single benign class, following (Frazer et al., 2021). Additionally, the authors applied Ensembl VEP (McLaren et al., 2016) to categorize variants by consequences. The original task from BEND benchmark is designed as a zero-shot classification task, but in this work, we applied it in a supervised classification setting. Further details on the evaluation procedure are provided in Appendix A.3

The current state-of-the-art performance on this benchmark is achieved by CADD v1.7, which reports an AUROC of 0.992 (Rentzsch et al., 2019). It is important to note that CADD is an aggregate scoring method that integrates multiple predictive features, some of which may incorporate signals from models trained on ClinVar data.

Coding Pathogenicity: The coding pathogenicity prediction task focuses on classifying genetic variants located within coding regions of the genome as either pathogenic (disease-causing) or benign (non-disease-causing). Coding regions are responsible for protein synthesis and are therefore central to understanding disease mechanisms, making this task a critical component of medical genomics and precision medicine.

For this task, we use a dataset derived from the AlphaMissense (Cheng et al., 2023), which filters and curates missense variants from ClinVar (release date: 2022-09-10). Variants were mapped to the canonical transcript, and those labeled as "Benign" or "Likely benign" in ClinVar were treated as benign, while those labeled as "Pathogenic" or "Likely pathogenic" were treated as pathogenic. Variants containing the term "splice" in their annotation were removed (29 variants), as the focus of this task is on missense mutations rather than splicing effects. Following this processing, a total of 100,796 missense variants with at least one review star were considered.

The final test dataset consists of 82,872 variants from 7,951 proteins, after removing those used for validation. Of these, 30,884 are pathogenic and 51,988 are benign. This dataset reflects real-world class imbalance but remains significantly more balanced than typical noncoding variant datasets, offering a robust benchmark for model evaluation. Moreover, coding variants are generally better annotated and more frequently subjected to functional validation, which contributes to higher data quality in this task.

Importantly, variants in coding regions frequently lead to missense mutations that directly alter protein structure or function. While traditional tools such as CADD (Rentzsch et al., 2019) have been widely used for this problem, novel GFM approaches, such as GPN-MSA, are beginning to surpass conventional methods in predictive performance.

In this study, we adopt the test set proposed in the AlphaMissense paper for both model training and evaluation to assess coding pathogenicity performance. AlphaMissense has AUROC of 0.940. Further details on the evaluation procedure are provided in Appendix A.3.

eQTL Prediction: This task involves predicting whether a genetic variant influences gene expression. The ability to predict expression-altering variants has far-reaching implications for understanding the genetic basis of traits and diseases. Accurate predictions can guide experimental validation, prioritize variants for functional studies, and inform the development of gene therapies.

The dataset used for this task was adapted from the Borzoi study (Linder et al., 2025) and consists of tissue-specific VCF files for 41 human tissues (filtered from the original 49 due to too few data samples in 8 tissues), publicly available at <https://console.cloud.google.com/storage/browser/borzoi-paper/ctl/eqtl>. The balanced dataset comprises a total of 97,922 variants, consisting of half positive eQTLs and an equal number of matched negative controls. The positive set includes variants with a posterior inclusion probability greater than 0.99 following SuSiE fine-mapping (Wang et al., 2021). To control for genomic proximity as a confounding factor, negative controls were specifically chosen to match the positive eQTLs in their distance to the transcription start site (TSS), following the procedure detailed in the original study. For the linear probing evaluation on Variant Benchmark, a separate logistic regression model was trained for each fold within each tissue. Performance was evaluated using an 11-fold cross-validation scheme. AUROC was calculated for each fold and subsequently averaged to yield a mean AUROC for each tissue. A final aggregate AUROC and standard deviation were then computed by taking a weighted average of the per-tissue results.

Expression Prediction: The dataset for this task was derived from DeepSea (Zhou & Troyanskaya, 2015). Variants with positive labels were obtained from the GRASP eQTL database (Leslie et al., 2014), lifted over from hg37 to hg38 using PyLiftOver (<https://pypi.org/project/pyliftover/>), and filtered to exclude conversion errors. To construct

a balanced dataset, each positive variant was paired with a randomly selected negative control from the original dataset, resulting in 156,162 variants in total.

To further investigate the role of TSS distance to variants, we created two additional datasets corresponding to <360 bp and <31 kbp negative control sets, as outlined in Figure 2g. For the <360 bp version, negative controls were restricted to variants within 360 bp of a positive sample, following the evaluation procedure in DeepSea. This subset included 57,457 negative controls, compared to 78,081 negatives in the default balanced dataset. For the <31 kbp version, we combined the <360 bp set with additional negatives within 710 bp, 6.3 kbp, and 31 kbp of positive variants, using the original DeepSea controls.

These additional configurations for datasets were designed to test whether BioFM performance relies on variant-TSS distance information encoded by annotation tokens. This process enables a clearer assessment of whether model performance reflects genuine learning of regulatory effects rather than dependence on distance-based signals.

sQTL Prediction: The splicing effect prediction task aims to determine whether a genetic variant influences alternative splicing patterns. Evaluation splicing effects is critical for understanding the mechanisms by which variants contribute to diseases, particularly for disorders caused by aberrant RNA processing. Existing methods primarily focus on predicting splicing sites, rather than the direct effects of specific variants on splicing, leaving a gap in accurate variant-level predictions.

The dataset for this task was obtained from the Borzoi study (Linder et al., 2025), comprising tissue-specific VCF files with positive sQTLs and matched negative controls (<https://console.cloud.google.com/storage/browser/borzoi-paper/qtl/sqtl>), following a similar procedure as for the eQTL dataset. In total, 43,028 variants were collected. Initial evaluations revealed that several tissue-specific folds contained very few variants, in some cases lacking either positive or negative samples. To address this limitation and ensure robust evaluation, tissues were grouped into broader categories (brain, hematopoietic, digestive, urogenital, adipose, endocrine, and others). Within each cluster, duplicate variants across tissues were removed to avoid redundancy (e.g., overlapping positive sets from multiple brain tissues). Variant counts for each tissue cluster are reported in Table 2. Evaluation was performed by computing AUROC for each fold within a cluster, followed by a weighted average across folds to obtain the final AUROC and standard deviation.

Tissue Cluster	Variant Count
Brain	2923
Hematopoietic	2281
Digestive	5624
Urogenital	2875
Adipose	2657
Endocrine	2901
Others	4280

Table 2. sQTL statistics

Ancestry Prediction: The ancestry prediction task aims to classify individuals into one of five superpopulations based on genomic sequence data. To construct the dataset, we started with the human reference genome (HG38) and generated consensus sequences for each individual in the 1000 Genomes Project (Consortium et al., 2015) by applying their corresponding mutations. For each individual, 11 genomic regions were selected, with each region comprising 32,000 nucleotides. These regions are sampled from alternate chromosomes (e.g., chromosomes 1, 3, 5, etc.), and their positions are fixed across all individuals in the dataset. Variability arises exclusively from the individual-specific mutations applied to these regions.

On average, the sequences differ from the reference at approximately 0.5% of positions, corresponding to around 33 variants per sequence. These variants include single nucleotide polymorphisms, insertions, and deletions. To ensure data quality, the selected regions are constrained such that all individuals in the 1000 Genomes dataset are fully represented in each region, i.e., there are no ambiguous or unknown bases, and the regions exhibit sufficient genetic variation across samples.

In the final dataset, the 11 regions are treated as independent samples, and each is assigned a superpopulation label corresponding to the individual from whom it was derived. This design results in a dataset that is 11 times larger than the number of individuals in the original 1000 Genomes dataset. To create a train-test split, we first split individuals into train and test, and then split each of 11 region sets by region individual label, so that no individual can have sequences in train and test set simultaneously.

For model training, each 32,000-nucleotide sequence is divided into smaller chunks to accommodate model-specific context length limitations (e.g., 6,144 for our BioFM, 8,192 for Evo2-1B-Base, and 12,288 for NT). Embeddings are extracted for each chunk, and a mean pooling operation is applied to obtain a single representation for the entire sequence. These aggregated embeddings are then used as inputs to a linear classifier to predict ancestry.

Common vs Rare Variants Classification: This task assesses whether model embeddings capture biologically conserved genomic contexts typical of authentic common variants. We randomly sampled 100,000 common variants (Minor Allele Frequency (MAF) > 0.05) from GnomAD (Chen et al., 2024) and, for each common variant, generated a synthetic negative control by randomly substituting a nucleotide within a ± 16 -nucleotide local sequence window. The resulting balanced dataset consists of exactly 200,000 single-nucleotide variants (SNVs), equally divided between genuine common variants and synthetic controls.

Generation of synthetic negative controls in a close proximity to common variants ensures that annotation tokens do not leak the information into the testing set. Moreover, during evaluation stage, both common variants and negative control variants are encoded with a variant token, therefore this task tests models ability to generate different embeddings for common and rare variants.

meQTL Prediction: This task involves predicting if a genetic variant influences nearby methylation rates. We downloaded meQTL dataset from GRASP (Leslie et al., 2014) which contains 52,418 datapoints. We filtered out duplicates by rsid, lifted variant coordinates to hg38 and obtained 25,496 variants (meQTLs) total. GRASP meQTL dataset itself is a compilation of meQTLs found by many studies, some of them are verified experimentally, others are fine-mapped. As this dataset contains only positive samples, to create negative controls, we randomly sampled 100,000 common variants (Minor Allele Frequency (MAF) > 0.05) from GnomAD (Chen et al., 2024), grouped them by chromosome and GC-content percentage with an precision of 1%, and for each meQTL variant we selected a random negative control from the corresponding group with the same chromosome and GC-content percentage. This procedure ensures that positive samples and negative controls are not easily distinguishable by GC-content around variant.

A.2. Modelling

ID	Token
0	D
1	N
2	A
3	T
4	G
5	C
6	[CLS]
7	[SEP]
8	[UNK]
9	[PAD]
10	[EOS]
11	[BOS]
12	[SPECIAL]
13	Ǻ
14	ǻ
15	ǿ
16	Ǿ
17	[START_CDS]
18	[END_CDS]
19	[START_TRANSCRIPT]
20	[END_TRANSCRIPT]
21	[START_EXON]
22	[END_EXON]

Table 3. BioToken Annotation Variant Vocabulary

A.2.1. BIOTOKEN

We experiment with three variants of BioToken to explore varying levels of genomic complexity. Each tokenizer incorporates unique aspects of genomic data, enhancing model understanding and performance. [Figure 1a](#) provides an overview of nucleotide-level points of interest and their integration into tokenization strategies. A brief description of the tokenizers is provided below: **BioToken-Base**: This baseline tokenizer maps each nucleotide (A, T, G, C) to a distinct token. It is commonly used in genomic foundation models ([Nguyen et al., 2024](#)) and offers a simple yet effective representation that preserves base-level positional information and facilitates preprocessing. However, this character-level encoding treats all genomic positions equivalently, ignoring biological context or inductive biases. Unlike natural languages, DNA sequences convey meaning not through syntactic rules, but through biochemical mechanisms, spatial chromatin structure, and regulatory interactions. As a result, modeling the genome purely as linguistic text overlooks its intrinsic organization and biological regularities.

BioToken-Variant: To capture individual-specific genetic variation, this tokenizer extends the BioToken-Base by incorporating variant information relative to the reference genome. For each individual from the Genome1000 dataset, variants within the sampled genomic regions are explicitly identified and encoded. Special symbols— $\check{A}$, $\check{T}$, $\check{G}$, $\check{C}$ —are used to represent single nucleotide variants and insertions. This design enhances the model’s ability to detect and learn from regions of genomic diversity.

BioToken-Annotation-Variant: Building upon the BioToken-Variant, this approach incorporates additional functional annotations crucial for understanding genomic patterns. Annotations include start and end of coding sequences, transcription regions, and exon regions providing context about regions critical to gene expression. These annotations enable the model to better capture biologically significant patterns. We use the GENCODE annotations version 38 ([Mudge et al., 2025](#)). We retain only CDS, transcript, and exon annotations, and remove exact duplicates based on annotation type and start/end coordinates. If two annotations of the same type share the same start position, both are included in sequence (e.g., two consecutive start_exon tokens).

It is important for the model to know the extent of alternative splicing of genes. The vocabulary for the annotation variant tokenizer is provided in [Table 3](#).

A.2.2. BIOFM

We utilize a decoder-only transformer architecture derived from Mistral ([Jiang et al., 2023](#)) as the foundation for our model. The architecture consists of 23 transformer layers with a total of 265M parameters. While we retained the grouped-query attention mechanism from the original Mistral architecture to improve inference efficiency, we disabled the sliding-window attention feature to allow the model to access the full input context without local attention constraints. To accommodate longer genomic sequences, we modified the Rotary Position Embeddings (RoPE) by increasing the θ parameter from 10^4 to 10^5 , following established approaches for extending context length in transformer models ([Xiong et al., 2023](#)). This modification enables the model to better handle positional information across extended DNA sequences that are typical in genomic applications. All remaining hyperparameters, including learning rate, batch size, and optimization settings, are detailed in [Table 4](#).

A.3. Evaluation

Variant Benchmark: For each task in Variant Benchmark, except ancestry prediction, we divide variants into 11 folds. Each fold has 21 chromosomes as training set and 2 chromosomes as testing set, except fold 0, which has chromosomes 21, 22, and X as testing set. For encoder models, we put variant in the center of context window, construct embeddings of reference sequence and mutated sequence and concatenate them. For decoder-only models, we construct forward mutated sequence which ends with variant, and reverse complement mutated sequence which also ends with variant. Both forward and reverse sequences have length of exactly half of evaluation context size. After that, we average forward and reverse reference sequences, and forward and reverse mutated sequences, and concatenate the resulting two vectors. This approach ensures that each model has access to the exact same amount of context information and also works around the fact that decoder-only models are causal.

After obtaining embeddings for each variant, we concatenate them into a feature matrix for a linear regression. We call it linear probing. We decided against full or parameter-efficient finetuning to mitigate the overfitting effects and effect of many

Parameter	Value
optimizer	AdamW
max_learning_rate	$7.2e - 4$
min_learning_rate	$3.2e - 5$
weight_decay	0.1
optimizer_momentum	$\beta_1, \beta_2 = 0.9, 0.999$
batch_size	64
context_size	6144
lr schedule	cosine with warmup
epochs	2
initializer_range	0.03
hidden_size	640
n_layers	23
n_heads	32
n_gqa	8

Table 4. Hyperparameters for BioFM 265M.

additional degrees of freedom coming from having many hyperparameter combinations to select from. Linear probing of model embeddings also allows us to isolate the quality of obtained representations. We used default settings in a sklearn package, but with number of steps for lbgfs method set to 5000. We report AUROC for each test set in each fold, and then obtain average AUROC for Figure 2a or median AUROC for Figure 1c and Figure 2g. It allows us to get more robust performance estimates for each model.

For comparison with Enformer, we use a PyTorch Enformer implementation from EleutherAI github.com/lucidrains/enformer-pytorch. Enformer first compresses each sequence into blocks using convolutions. Following original evaluation approach in (Avsec et al., 2021), we extract three hidden states for blocks centered about variant of interest, average them into one hidden state over sequence dimension and use it for the linear probing. Additionally, we ran an experiment where we used only block with the variant, but results did not change. We evaluated Enformer on expression prediction task from Variant Benchmark using context sizes of 12K, 24K, 48K, and 96K. Performance of Enformer saturated for 48K and 96K nucleotides, therefore we did not evaluate it on 192K context.

For comparison with Splice Transformer, we use an official implementation from authors at github.com/ShenLab-Genomics/SpliceTransformer. We always use the maximum available context size of 8K nucleotides, centered about variant of interest, extract embeddings for this variant only, and use linear probing on top.

For comparison with Evo2-1B and Evo2-7B-Base models, we use embeddings from layer 19 and layer 24, respectively. We selected these layer indices, because embeddings from last layer performed significantly worse and in (Brix et al., 2026) these layer embeddings were generally among the best for coding pathogenicity prediction in BRCA1 and BRCA2 tasks. We also use mean pooling for all results shown in Figure 2a. It consistently shows better performance compared to last token pooling. We also evaluated context sizes (sequence length) for both last token pooling and mean pooling from 1024 to 8192, and on average, mean pooling with sequence length of 1024 works better. Therefore, to create a maximally favourable conditions for Evo2 set of models, we use specific layer embeddings, mean pooling, and context size of 1024.

GUE Evaluation: We use the existing train-validation-test splits on the original GUE dataset for finetuning purposes. For each model, we ran a sweep through 4 values of learning rate and 2 values of maximum epochs. The validation dataset is used for model selection, and the test scores corresponding to model with the best validation score is reported. As stated in Vishniakov et al., we find that max-pooling performs better than cls/last token pooling during our initial experiments (Vishniakov et al., 2026). Our finetuning configuration is provided in Table 6.

Dart Evaluation: We selected two tasks from Dart-Eval benchmark (Patel et al., 2024), namely African caQTL prediction and Yoruba dsQTL prediction. As original datasets are in HG37 assembly, we lifted them to HG38 assembly and filtered out variants with reference allele mismatch. After liftover, number of variants in African caQTL dataset is 79,026, and number

Metric	Caduceus	HyenaDNA	NT-500m-v2-Multispecies	GenaLM	DNABERTv2	BioFM
EM Mean	0.563	0.497	0.588	0.493	0.648	0.628
H3	0.721	0.727	0.765	0.699	0.806	0.773
H3K14ac	0.513	0.406	0.565	0.455	0.653	0.604
H3K36me3	0.583	0.472	0.632	0.473	0.641	0.640
H3K4me1	0.459	0.412	0.552	0.380	0.537	0.537
H3K4me2	0.422	0.331	0.381	0.321	0.504	0.442
H3K4me3	0.445	0.341	0.444	0.343	0.545	0.572
H3K79me3	0.657	0.588	0.645	0.611	0.721	0.688
H3K9ac	0.513	0.515	0.601	0.517	0.655	0.607
H4	0.775	0.758	0.815	0.723	0.808	0.804
H4AC	0.544	0.424	0.480	0.410	0.606	0.612
PD Mean	0.720	0.780	0.911	0.772	0.802	0.831
all	0.844	0.841	0.931	0.828	0.778	0.869
notata	0.923	0.915	0.949	0.915	0.940	0.944
tata	0.394	0.584	0.854	0.573	0.687	0.680
TFH Mean	0.619	0.588	0.661	0.580	0.700	0.663
TF0	0.641	0.611	0.663	0.668	0.708	0.705
TF1	0.692	0.662	0.718	0.661	0.740	0.709
TF2	0.585	0.596	0.634	0.561	0.660	0.659
TF3	0.488	0.452	0.539	0.316	0.598	0.553
TF4	0.690	0.620	0.750	0.692	0.792	0.691
CPD Mean	0.541	0.631	0.752	0.607	0.694	0.695
all.1	0.621	0.633	0.719	0.599	0.663	0.652
tata.1	0.350	0.596	0.822	0.561	0.730	0.741
notata.1	0.653	0.664	0.715	0.660	0.688	0.691

Table 5. Full GUE Results.

config	value
optimizer	AdamW
learning_rate	3e-5, 5e-5, 7e-5, 1e-4
weight_decay	0.01
warmup_steps	40
optimizer_momentum	$\beta_1, \beta_2 = 0.9, 0.999$
batch_size	32
lr schedule	cosine
epochs	10 / 20

Table 6. Configuration for GUE fine-tuning.

of variants in Yoruba dsQTL dataset is 27,373. Prevalence of caQTLs in African dataset is 0.086, and prevalence of Yoruba dsQTLs is 0.02. We used the same evaluation procedure as for our own Variant Benchmark, splitting data into 11 folds by chromosome and performing linear probing on top of every model embeddings afterwards. We used last layer embeddings and maximum context size up to 12K for every model, except Evo2 models, where we used context size of 1K and layers 19 and 24 for Evo2-1B-Base and Evo2-7B-Base, respectively. We chose to report AUROC to be consistent with choice of other benchmarks metrics.

Trait Gym Evaluation: Trait Gym has two variant-related tasks, called Complex traits prediction and Mendelian traits prediction. Both tasks include binary classification of variants which can influence a trait. The detailed variant selection procedure is described in (Benegas et al., 2025b). Complex traits dataset has 10K variants, and Mendelian traits dataset has 3380 variants. We are using the same evaluation protocol, as in (Benegas et al., 2025b). More specifically, for complex traits dataset, we split data into 22 folds, one fold per chromosome. We use linear probing with Elastic-Net regularization using GridCV search for the regularization parameter C from scikit-learn library. The C parameter grid is from 10^{-8} to 10^{10} , similar to (Benegas et al., 2025b). For the selection of the best value of C , we split training part of the dataset into 5 folds with train and validation parts and select C based on validation part of the dataset. Final reported metric is a weighted average AUROC, calculated on test splits of the dataset. We used last layer embeddings and maximum context size up to 12K for every model, except Evo2 models, where we used context size of 1K and layers 19 and 24 for Evo2-1B-Base and Evo2-7B-Base, respectively.

Promoter Generation: We use a high-quality experimental dataset EPDNew (Meylan et al., 2020) for the Supervised Fine-Tuning (SFT). It contains 20,172 promoters for 9K genes. We split this dataset into 90% train genes and 10% test genes. After constructing gene sequences and accounting for reverse strands, we annotate each gene sequence with relevant metadata corresponding to the promoter dataset. Promoters are extracted using a $-249, +50$ bp window around the transcription start site, following the approach used in the GUE dataset (Zhou et al., 2024). Additionally, we collect gene-specific information, including gene symbol, start, and end positions, to provide better contextual representation of the generated promoters. We fine-tune our pre-trained model for promoter generation using the SFTTrainer from the TRL library (von Werra et al., 2020). Our training data follows the structured response template: `<gene_sequence> P <promoter_sequence>`, which serves as inputs for the SFTTrainer.

Ablation, Scaling, Representation Analysis, and Attention Specialization: We conducted detailed analyses to better understand BioFM’s internal representations and the influence of training scale on downstream task performance.

For representation analysis, we constructed two-component UMAP embeddings of variant representations from the smaller 134M-parameter BioFM model, which contains 15 layers. Sequences used for embedding were fixed at 1024 nucleotides in length for both forward and reverse strands, with each variant consistently positioned at the sequence end. We averaged embeddings from forward and reverse strands to obtain the final variant representation. UMAP embeddings were computed using the Python umap package, with parameters set to minimal distance 0.05 and number of neighbors 10. To assess biological distinctions captured by these embeddings, we subsampled 500 variants for each Ensembl VEP consequence type from pathogenicity datasets within the Variant Benchmark.

To evaluate how downstream linear probing performance scales with the number of training tokens and finetuning samples,

config	value
train_batch_size	16
eval_batch_size	16
learning_rate	1e-3
weight_decay	0.01
warmup_steps	40
lr schedule	cosine
epochs	10
max_seq_length	6144

Table 7. Configuration for SFT step.

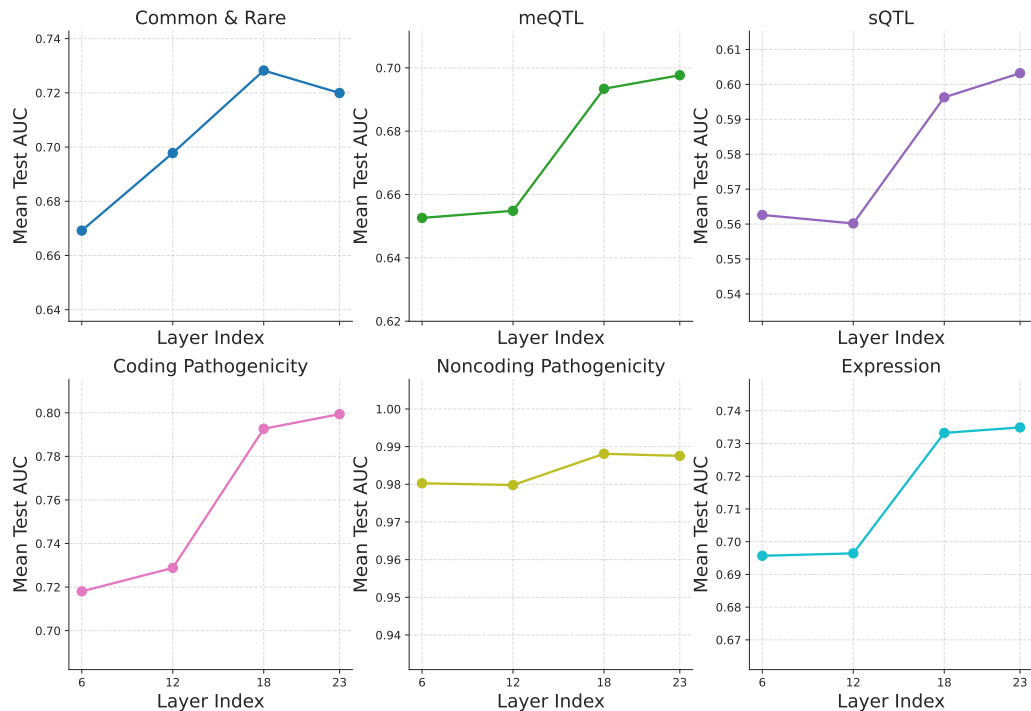


Figure 6. Different layer embeddings performance of BioFM 265M.

we subsampled only the training sets from each fold across seven tasks in the Variant Benchmark. The test sets remained constant throughout these evaluations. We considered five model checkpoints trained on varying amounts of genomic data, ranging from 8.6B tokens (equivalent to approximately 2.5 genomes) to 103.1B tokens (approximately 35 genomes). For each subsampled training dataset, we computed the average AUROC across the full test sets of each task, thereby systematically quantifying the relationship between pretraining scale and downstream prediction accuracy.

Finally, to investigate attention specialization, we analyzed attention maps using a fixed 64-nucleotide sequence containing a small exon and coding sequence (CDS) at its center. We examined attention heads specifically selected for their hypothesized biological relevance: annotation and start/variant token heads from the initial layer (layer 0), and periodic and coding/noncoding heads from the final layer. This analysis aimed to provide mechanistic insights into how BioFM learns biologically meaningful patterns at the nucleotide level.

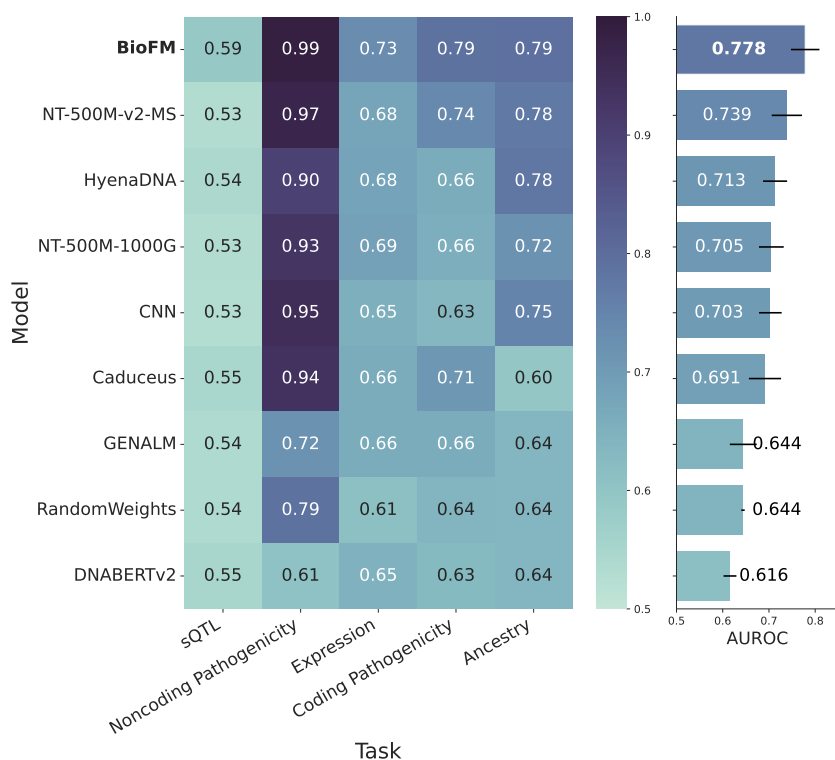


Figure 7. Variant Benchmark linear probing GFM comparison using context length of 1K. RandomWeights model correspond to the untrained version of BioFM 134M, which essentially produces random embeddings.

A.4. Pseudocode for BioFM Inference and Embedding Extraction

This pseudocode illustrates how BioFM-Eval Python library processes VCF files to generate biologically informed embeddings using the BioFM model. It includes variant annotation, context-aware sequence construction, tokenization, and final embedding extraction.

Algorithm 1 Variant Embedding Extraction Pipeline in BioFM-Eval

Input : VCF file $\mathcal{V}$, reference genome $\mathcal{G}$, gene annotations $\mathcal{A}$, pretrained model $\mathcal{M}$, tokenizer $\mathcal{T}$
Output : Embeddings $\mathbf{E} \in \mathbb{R}^{N \times 2d}$ for N valid variants

 Initialize Annotator using $(\mathcal{A}, \mathcal{G})$

 Load pretrained model $\mathcal{M}$ and tokenizer $\mathcal{T}$

 Embed $\leftarrow$ Initialize Embedder with $(\mathcal{M}, \mathcal{T})$
foreach variant v_i in VCF file $\mathcal{V}$ **do**

Extract reference context:

 $s_{up}^{ref} \leftarrow$ upstream reference sequence before v_i
 $s_{down}^{ref} \leftarrow$ downstream reference sequence after v_i

Apply mutation to get mutated sequences:

 $s_{up}^{mut} \leftarrow$ mutated upstream sequence ending at v_i
 $s_{down}^{mut} \leftarrow$ mutated downstream sequence starting at v_i

Reverse complement downstream sequences:

 $s_{down}^{ref} \leftarrow \text{RevComp}(s_{down}^{ref})$
 $s_{down}^{mut} \leftarrow \text{RevComp}(s_{down}^{mut})$

Compute embeddings:

 $e_{up}^{ref} \leftarrow \text{Embed}(s_{up}^{ref})$
 $e_{down}^{ref} \leftarrow \text{Embed}(s_{down}^{ref})$
 $e_{up}^{mut} \leftarrow \text{Embed}(s_{up}^{mut})$
 $e_{down}^{mut} \leftarrow \text{Embed}(s_{down}^{mut})$

Average embeddings:

 $e_{ref} \leftarrow \frac{1}{2}(e_{up}^{ref} + e_{down}^{ref})$
 $e_{mut} \leftarrow \frac{1}{2}(e_{up}^{mut} + e_{down}^{mut})$

Concatenate to form final embedding:

 $e_i \leftarrow [e_{ref} || e_{mut}]$

 Append e_i to embedding matrix $\mathbf{E}$
return $\mathbf{E}$

A.5. Statistical Tests

We performed a two-sided Wilcoxon signed-rank test (exact method; SciPy library (Virtanen et al., 2020)) to evaluate whether the differences in AUC between BioFM and Enformer across 11 cross-validation folds were symmetrically distributed around zero. This non-parametric test does not require normality but assumes symmetry of the differences around the median. BioFM significantly outperformed Enformer (Wilcoxon signed-rank statistic $T=1$, and $p=0.002$).

A.6. Representation Analysis

We also evaluated BioFM’s ability to utilize large-scale pretraining well by analyzing its sample efficiency with linear probing on variant benchmark tasks. We expected the downstream task performance differences to be larger when the number of finetuning samples is low. In agreement with this hypothesis, we observed that the difference between undertrained (8.6B, 17.2B) and trained models (68.7B, 103.1B) is particularly pronounced when the number of finetuning samples is low (Figure 8a). The difference reduces when we increase number of fine-tuning samples for linear probing. This observation illustrates that there is a clear benefit to scaling training data size, with the law of diminishing returns eventually becoming apparent.

Lastly, we probed BioFM’s internal attention patterns to explore how biologically-informed annotations and variant tokens

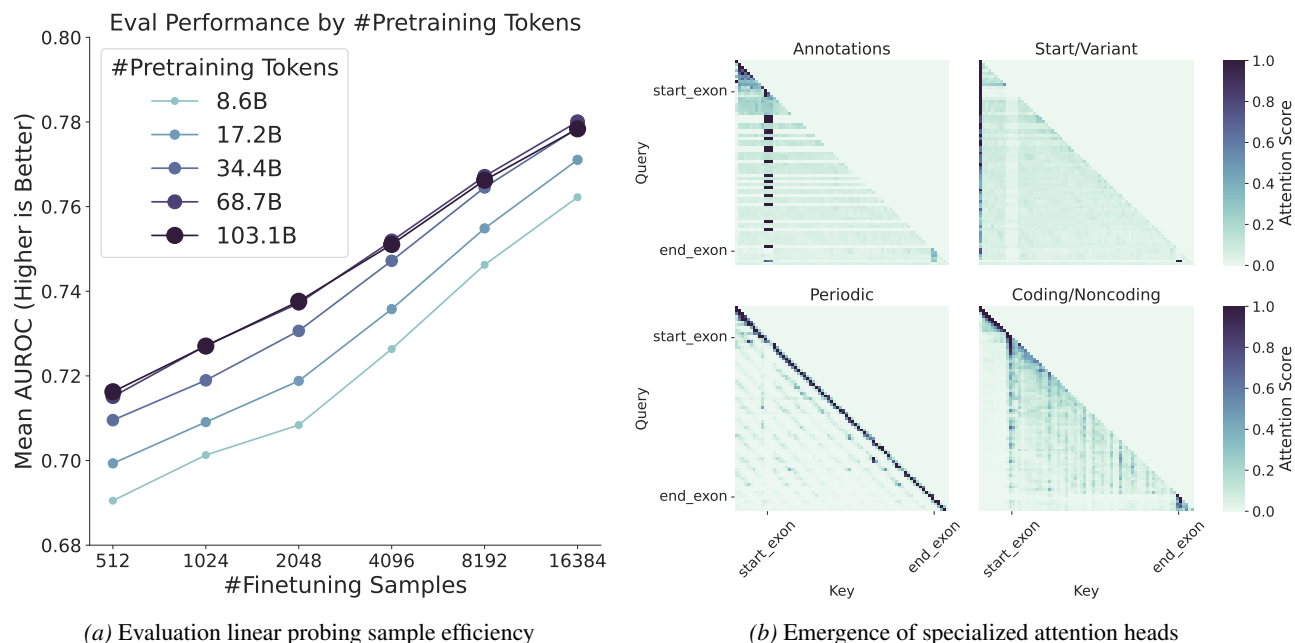


Figure 8. Detailed analysis shows that biologically-informed tokenization using BioToken in BioFM significantly enhances computational efficiency and attention maps show biologically meaningful patterns. **a)** Evaluation of linear probing efficiency highlights BioFM’s sample efficiency and scalability, benefiting significantly from increased genomic training data while achieving superior downstream performance. **b)** Visualization of attention maps provides mechanistic insight into specialized attention heads, clearly demonstrating BioFM’s ability to selectively attend to biologically significant genomic annotations and regions, further validating the functional interpretability and efficacy of the BioToken strategy.

affect representation learning at a mechanistic level (Figure 8b). We hypothesized that specific attention heads may be specializing in processing annotation and variant tokens based on inherent sequence differences between coding and noncoding regions. This expectation aligns with the biological observation that the probability distribution of DNA k-mers differs between coding and noncoding regions—coding regions exhibit constrained sequence variation, while noncoding regions often contain repetitive elements. Attention mechanisms should thus help the model distinguish between these contexts by learning distinct probability distribution patterns.

To test this hypothesis, we selected a 64-base sequence that includes a short exon sequence in the middle, flanked by `<start_exon>` and `<end_exon>` tokens, with a variant token at the end. Visualization of attention maps for this sequence revealed distinct patterns among attention heads (Figure 8b). Some attention heads were dedicated to annotation tokens, while others focused on sequence start and variant tokens, potentially aiding in the recognition of contextual boundaries and mutation sites. Certain other attention heads exhibited periodic patterns, which may reflect the repetitive structure of noncoding regions, often associated with regulatory or structural genomic elements. Additionally, some attention heads segregated exonic and intronic regions, indicating a learned distinction between protein-coding and noncoding sequences. These findings indicate that the model recognizes biologically meaningful sequence features, offering insights into how BioToken benefits BioFM in encoding genomic information.

A.7. Evaluation of GFMs with Max Context

In addition to the evaluation of all models up to minimum of (12K nucleotides, their maximum supported sequence length), we pushed Caduceus and HyenaDNA to the limit permitted by our hardware, which is 131K nucleotides and corresponds to what Caduceus was trained for at the end of their context length extension phase. Caduceus showed significant improvement only for the expression prediction, while for other 6 tasks performance stayed the same Table 8. HyenaDNA displayed the same pattern, AUROC for expression prediction improved from 0.68 to 0.72. Performance for sQTL prediction improved from 0.55 to 0.58 AUROC. On the other hand, AUROC for classifying common and rare variants decreased from 0.68 to 0.66. BioFM, NT-500M-v2-MS, Evo2-1B-Base, and NT-500M-1000G were already evaluated on every possible context size, while Evo2-7B showed significant decrease in performance from 1K nucleotides to 8K nucleotides, therefore we

decided not to extend its context size even further.

A.8. Annotations Effect

To check if annotations can help other models improve their performance as well, we performed additional set of experiments on Variant Benchmark. To add annotation features, we simply counted number of coding region, exon, transcript starts and ends in the window of 128bp or 12K bp for each variant and concatenated this 6-point vector with model embedding for each variant. Also, we changed a few things in our evaluation protocol to facilitate using these counted annotation features. First, we removed ancestry prediction task, because it is completely independent from annotations (we always use the exact same genome regions for every individual, so the annotations will also be exactly the same). Second, we substituted linear probing with XGBoost for easier integration of annotation features into embeddings.

Notably, Evo2 models performed slightly worse, BioFM showed the same performance, NT improved, while smaller HyenaDNA and Caduceus improved noticeably up to AUROC 0.777. We additionally tested Caduceus with XGBoost but without annotation features and found out that half of the Caduceus improvement can be attributed to power of downstream predictor, XGBoost-only AUROC is 0.753. Overall, results confirm the hypothesis that bigger models trained on trillions of tokens have to learn some annotations themselves, BioFM is able to perfectly use annotation features as designed, smaller models do benefit from feature engineering and more powerful downstream predictor. Additionally, we varied the window in which we calculate annotation counts up to 12K bp and found out that performance of smaller models and NT-500M-v2-MS did improve a little further, while every other model performed the same. BioFM also shows insensitivity to the choice of downstream predictor, proving that representations it learned are easier to use in general.

A.9. Pooling and Context Length Summary

In general, we tried to give each GFM the best chance in our benchmark by evaluating different context sizes. We show that using maximum context length for each model does not necessarily lead to best performance. The same fact was established in (Brixi et al., 2026), where extensive context size tuning on test set for BRCA1 and BRCA2 mutation effect prediction did yield context size of 128 as the best one. Moreover, embeddings from intermediate layers were significantly better. Similar phenomena was observed in (Dalla-Torre et al., 2025), where intermediate layer embeddings were generally better for region classification tasks. Therefore, we were forced to evaluate multiple context sizes for every model. Summary of all our decisions is in Table 10.

For BioFM we simply used the variant token embedding and maximum context size of 6144 nucleotides. BioFM is a decoder-only model, therefore to incorporate information about nucleotides after the variant token, we run BioFM for the reverse complement sequence which ends with variant token. In total, we use 12288 nucleotides. The exact same procedure is applied to HyenaDNA, because it is also a decoder-only model.

For NT-500M-v2-MS we use the maximum context size of 12288 nucleotides and take the embedding of a variant token which is in the middle of a sequence. The same procedure, but with smaller context size of 6144 nucleotides is performed for NT-500M-1000G. We rigorously confirmed that the maximizing context size for NT family models maximizes their performance.

For Caduceus we use 12288 nucleotides from the forward strand, centered about variant, the same as for NT-500M-v2-MS. Caduceus is a bi-directional encoder, therefore variant token embedding will have information about the whole context

Task	Caduceus (12K)	Caduceus (131K)	HyenaDNA (12K)	HyenaDNA (131K)
sQTL	0.59	0.59	0.55	0.58
meQTL	0.73	0.74	0.65	0.66
eQTL	0.68	0.68	0.65	0.65
Noncoding Pathogenicity	0.95	0.95	0.91	0.91
Expression	0.67	0.71	0.68	0.72
Common and Rare	0.69	0.69	0.68	0.66
Coding Pathogenicity	0.72	0.72	0.67	0.67

Table 8. Pushing Caduceus and HyenaDNA to the context size limit, metric is always AUROC, context size is in brackets

window. We use PS variant of the Caduceus, because it is better than PH. We again tested the Caduceus for different context sizes and found out that 12288 nucleotides allows us to saturate its performance.

For Evo2-1B-Base and Evo2-7B models, we report results for 1024 context size and mean pooling. We use 512 nucleotides from the forward strand ending with variant token, and reverse complement sequence of 512 nucleotides starting from variant token. If we increase context size **or** change pooling to the variant token, average performance of Evo2-1B-Base drops significantly. Therefore, we again make our best attempt to elicit as much performance as we can from any model in our benchmark.

A.10. Comparison with Borzoi

We run Borzoi on eQTL and sQTL prediction tasks using our linear probing protocol same as for Enformer. We used 196K context, averaged three central bins of Borzoi embeddings to get a final embedding (as recommended by authors of Enformer and Borzoi), and also averaged the final embeddings of 4 Borzoi replicas. Borzoi achieves AUROC of 0.689 for eQTL compared to BioFM 0.699 and AUROC of 0.638 for sQTL, compared to BioFM 0.683. In the original paper ([Linder et al., 2025](#)), eQTL and sQTL results were slightly higher, because authors used Random Forest instead of linear probing, and more importantly, random split into 8 folds, while we used the chromosome-based split. Random splits are well-known to inflate models predictive performance in genomics, e.g. (TargetFinder).

Model	Baseline AUROC	Window 128bp	Window 12K bp
BioFM	0.780	0.786	0.786
NT-500M-v2-MS	0.720	0.731	0.748
Evo2-7B	0.734	0.714	0.716
Caduceus	0.719	0.764	0.777
HyenaDNA	0.687	0.724	0.738
Evo2-1B-Base	0.704	0.706	0.708

Table 9. Effect of adding count of CDS, Exon, Transcript annotations as separate features

Model	Pooling	Context Size
BioFM	Variant Token	6144 Forward and 6144 Reverse
NT-500M-v2-MS	Variant Token	12288 Forward
Evo2-7B	Mean	512 Forward + 512 Reverse
Caduceus	Variant Token	12288 Forward
HyenaDNA	Variant Token	6144 Forward and 6144 Reverse
Evo2-1B-Base	Mean	512 Forward + 512 Reverse
NT-500M-1000G	Variant Token	6144 Forward

Table 10. Pooling and context size summary